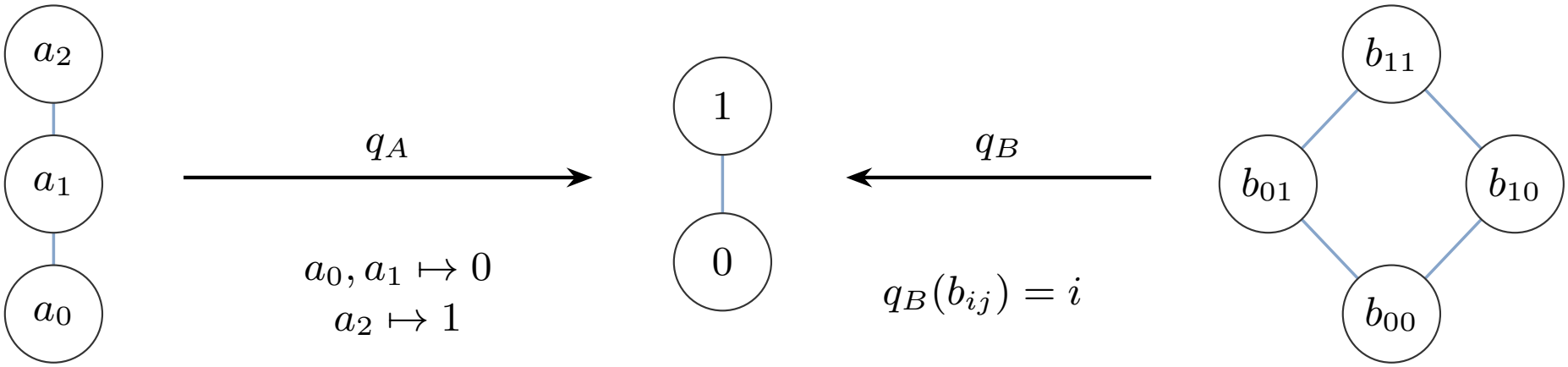
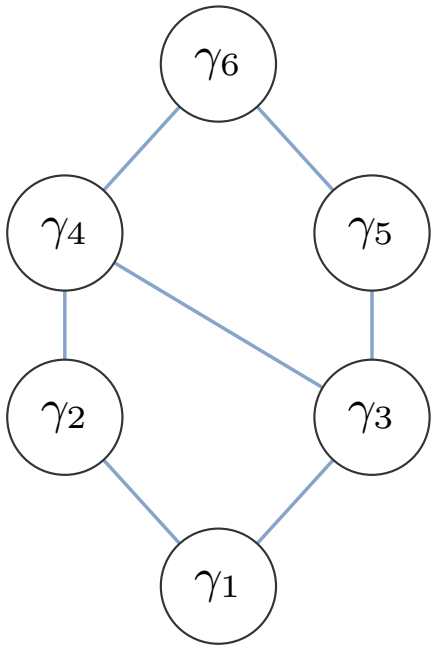
$L_A = C_3$ $K = C_2$ $L_B = C_2 \times C_2$



$$P_K = \{(a,b) : q_A(a) = q_B(b)\} \quad \textbf{Keep equal-image pairs only}$$
$$|P_K| = |q_A^{-1}(0)||q_B^{-1}(0)| + |q_A^{-1}(1)||q_B^{-1}(1)| = 2 \cdot 2 + 1 \cdot 2 = 6$$



Common image 0

$\gamma_1 = (a_0, b_{00}), \quad \gamma_2 = (a_0, b_{01})$

$\gamma_3 = (a_1, b_{00}), \quad \gamma_4 = (a_1, b_{01})$

Common image 1

$\gamma_5 = (a_2, b_{10}), \quad \gamma_6 = (a_2, b_{11})$

Local lattices are non-isomorphic; the nonempty pullback is a lattice